Teaching Note: The Hartmann Legacy

Vertraulich — nur für Lehrende

BUA3 --- Business Finance, THWS

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1 Learning Objectives

After completing this case session, students will be able to:

1. Apply the compound interest formula to calculate future values and isolate unknown interest rates algebraically.
2. Use the Fisher equation to distinguish nominal from real returns and identify negative real rates.
3. Value a finite annuity stream using PVIFA and size a loan using the CRF; interpret the limit case (perpetuity).
4. Build complete amortization tables for all three standard loan types and compare them on total cost and liquidity profile.
5. Price a bond as PV(coupons) + PV(par) and explain the discount/premium logic; estimate YTM via linear interpolation.

6. Apply the Gordon Growth Model to value a dividend-paying stock and assess over/undervaluation.

2 Synopsis

The Hartmann Family Office has inherited four unresolved financial questions from the retiring founder. A newly returned daughter, Lena, must work through each in sequence: valuing a matured deposit in real terms, deciding whether a commercial property is worth its asking price and how to finance it, comparing three loan structures for a renovation project, and evaluating two portfolio positions (a bond and a stock). The case integrates Chapters 1–4 of BUA3 in a single coherent narrative.

3 Theoretical Framework

Task	Chapter	Core Concept
Aufgabe 1	1	Compound interest, Future Value, Fisher Equation, algebraic rate isolation
Aufgabe 2	2	PVIFA, CRF, perpetuity, Gordon model precursor
Aufgabe 3	3	Three loan types, amortization mechanics, liquidity vs. total cost trade-off
Aufgabe 4	4	Bond valuation (discount/premium), YTM interpolation, DDM/Gordon Growth Model

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4 Full Model Solutions

4.1 Aufgabe 1 — Time Value & Real Returns

4.1.1 a) Future Value of the Deposit

$$FV = PV \cdot (1+i)^n = 500,000 \cdot (1.04)^5$$

$(1.04)^5 = 1.21665$

$$\boxed{FV = 500,000 \times 1.21665 = \textbf{€608,326.45}}$$

4.1.2 b) Real Interest Rate

Approximation (Fisher shortcut):

$$r \approx i_{nom} - \pi = 4.0\% - 5.5\% = \textbf{-1.5\%}$$

Exact Fisher Equation:

$$r = \frac{1 + i_{nom}}{1 + \pi} - 1 = \frac{1.04}{1.055} - 1 = 0.98578 - 1 = \textbf{-1.422\%}$$

Interpretation: Despite earning 4% nominally, Georg's purchasing power *fell* by approximately 1.4% per year. The real return was negative — the deposit grew in euros, but those euros bought less each year than the ones he had deposited. This mirrors the 2019–2024 European inflation shock discussed in Chapter 1 (Turkey case analogy).

> **Common error:** Students often compute inflation – nominal rate = +1.5%, reversing the sign. The real rate is what remains *after* deducting inflation from the nominal rate. A negative real rate means you are losing purchasing power even while the account balance grows.

Time: ~15 min

4.1.3 c) Required Annual Compound Rate

Solve $FV = PV \cdot (1 + i)^n$ for i:

$$280,000 = 200,000 \cdot (1 + i)^6$$

$$(1 + i)^6 = \frac{280,000}{200,000} = 1.4$$

$$i = 1.4^{1/6} - 1 = \sqrt[6]{1.4} - 1$$

$$\boxed{i \approx 5.77\% \quad \text{p.a.}}$$

> **Common error:** Students often forget to subtract 1 at the end, reporting i = 1.0577. Reinforce the algebraic step: the factor (1+i)^n must be isolated *before* taking the root.

Time: ~15 min

4.2 Aufgabe 2 — Annuities & Perpetuities

4.2.1 a) Present Value of Rental Stream

$$PVIFA(6\%, 15) = \frac{1 - (1.06)^{-15}}{0.06}$$

$(1.06)^{15} = 2.39656$, so $(1.06)^{-15} = 0.41727$

$$PVIFA = \frac{1 - 0.41727}{0.06} = \frac{0.58273}{0.06} = \mathbf{9.7122}$$

$$PV = 96,000 \times 9.7122 = \boxed{\textbf{€932,376}}$$

Common error: Students forget to take the reciprocal when computing the discount factor from the growth factor. Drill: $(1.06)^{-15} = 1/(1.06)^{15}$ — the minus sign is a reciprocal, not a subtraction.

Time: ~15 min

4.2.2 b) Investment Decision

The PV of the rental stream (€932,376) **exceeds** the asking price (€900,000) by approximately **€32,376**. The property is worth more than it costs → **recommend purchase.**

Discussion prompt if class is stuck: "If you were offered a promise to pay you €96,000 per year for 15 years, and you could earn 6% elsewhere, how much would you pay for that promise today?" — This frames it as a pure valuation question, removing the property context.

Time: ~5 min

4.2.3 c) Annual Mortgage Payment

$$CRF(6\%, 15) = \frac{0.06 \times (1.06)^{15}}{(1.06)^{15} - 1} = \frac{0.06 \times 2.39656}{2.39656 - 1} = \frac{0.14379}{1.39656} = 0.10296$$

$$\text{Annual payment} = 800,000 \times 0.10296 = \boxed{\textbf{€82,370 p.a.}}$$

Note: CRF = 1/PVIFA. Both routes give the same answer — accept either approach.

Time: ~10 min

4.2.4 d) Perpetuity Value

$$PV_\infty = \frac{C}{i} = \frac{96,000}{0.06} = \boxed{\textbf{€1,600,000}}$$

The perpetuity value (€1.6m) is **€667,624 higher** than the 15-year annuity value (€932,376). This large gap reveals that years 16 to ∞, even though heavily discounted, still contribute significant value when aggregated. Conversely, it illustrates how much value is "front-loaded" into the first 15 years — the annuity already captures 58% of the perpetuity's value.

Board note: Write both values side by side and ask: "Where did the other €668k go?" Students often underestimate how much a finite term cuts value.

Time: ~10 min

4.3 Aufgabe 3 — Loan Structures & Amortization

4.3.1 a) Option A — Annuity Loan

$$CRF(5\%, 4) = \frac{0.05 \times (1.05)^4}{(1.05)^4 - 1} = \frac{0.05 \times 1.21551}{0.21551} = \frac{0.060776}{0.21551} = 0.28201$$

$$\text{Annual payment} = 240,000 \times 0.28201 = \mathbf{€67,682.84}$$

Year	Opening Balance (€)	Annuity (€)	Interest (€)	Principal (€)	Closing Balance (€)
1	240,000.00	67,682.84	12,000.00	55,682.84	184,317.16
2	184,317.16	67,682.84	9,215.86	58,466.98	125,850.18
3	125,850.18	67,682.84	6,292.51	61,390.33	64,459.85
4	64,459.85	67,682.84	3,222.99	64,459.85	0.00
Total		**€270,731.36**	**€30,731.36**	**€240,000.00**	

Common error (Scissors Effect): Students sometimes reverse the direction — they expect interest to *rise* as payments progress. Correct by pointing to Year 1 vs. Year 4: interest falls from €12,000 to €3,223 because the outstanding principal is shrinking.

Time: ~20 min

4.3.2 b) Option B — Constant Amortization Loan

$$T = \frac{240,000}{4} = \mathbf{€60,000 \ \ per\ year}$$

Year	Opening Balance (€)	Total Payment (€)	Interest (€)	Principal (€)	Closing Balance (€)
1	240,000.00	72,000.00	12,000.00	60,000.00	180,000.00
2	180,000.00	69,000.00	9,000.00	60,000.00	120,000.00
3	120,000.00	66,000.00	6,000.00	60,000.00	60,000.00
4	60,000.00	63,000.00	3,000.00	60,000.00	0.00
Total		**€270,000.00**	**€30,000.00**	**€240,000.00**	

Time: ~15 min

4.3.3 c) Option C — Bullet Loan

$$\text{Interest (constant)} = 240,000 \times 0.05 = \mathbf{€12,000 \ \ per\ year}$$

- Years 1–3: **€12,000** (interest only; principal untouched)
- Year 4: **€12,000 + €240,000 = €252,000** (interest + full principal bullet)
- **Total paid: €288,000 | Total interest: €48,000**

Time: ~5 min

4.3.4 d) Comparison Table & Recommendation

Loan Type	Total Paid (€)	Total Interest (€)	Year 1 Outflow (€)
Option A — Annuity	**270,731.36**	30,731.36	**67,682.84**
Option B — Constant Amortization	270,000.00	**30,000.00**	72,000.00
Option C — Bullet	288,000.00	48,000.00	**12,000.00**

Recommendation for Lena: The property will be vacant in Year 1, meaning no rental income to service the debt. The Bullet Loan (Option C) minimises the Year 1 cash outflow at only €12,000, preserving liquidity during the renovation phase — at the cost of €18,000 more in total interest compared to Option B. This is the classic CFO "matching" logic from Chapter 3: match the loan structure to the asset's cash flow profile. The bullet structure matches a temporarily cash-flow-negative investment. Alternative: Option A (annuity) is a reasonable middle ground if the family expects rental income to resume from Year 2.

Discussion trap: Some students will instinctively recommend Option B because it has the lowest total interest cost. Push back: "Can Lena afford €72,000 in Year 1 with zero rental income?" Force the liquidity vs. total cost trade-off.

Time: ~10 min

4.4 Aufgabe 4 — Bond & Stock Valuation

4.4.1 a) Bond Valuation

$$P_0 = C \cdot PVIFA(YTM, n) + F \cdot (1 + YTM)^{-n}$$

With $C = 40$, $YTM = 6\%$, $n = 6$, $F = 1,000$:

$$PVIFA(6\%, 6) = \frac{1 - (1.06)^{-6}}{0.06} = \frac{1 - 0.70496}{0.06} = \frac{0.29504}{0.06} = 4.9173$$

$$\text{PV of coupons} = 40 \times 4.9173 = \text{€}\mathbf{196.69}$$

$$\text{PV of par value} = 1,000 \times (1.06)^{-6} = 1,000 \times 0.70496 = \text{€}\mathbf{704.96}$$

$$\boxed{P_0 = 196.69 + 704.96 = \text{€}\mathbf{901.65}}$$

The bond trades at a **discount** (below par). **Why:** The bond pays a coupon of 4%, but the market now demands 6% for comparable risk. No rational investor pays par for an asset yielding 4% when they can earn 6% elsewhere. The price must fall until the combination of below-market coupons *plus* the capital gain from buying below par exactly equals a 6% total return. The €98.35 discount (€1,000 – €901.65) is that compensation mechanism.

> **Common error:** Students confuse coupon rate (fixed, printed on the bond) with YTM (set by the market, varies daily). Reinforce: the coupon rate is the past; the YTM is the present.

Time: ~20 min

4.4.2 b) YTM Back-Calculation (Linear Interpolation)

Given: F = €1,000, c = 5% (C = €50), n = 5 years, market price = €980.

Step 1: Bracket with two trial rates.

At $r_1 = 5\%$: Since coupon = YTM, bond trades at par $\longrightarrow P(5\%) = €1,000.00$

At $r_2 = 6\%$:

$$P(6\%) = 50 \times PVIFA(6\%, 5) + 1,000 \times (1.06)^{-5}$$

$$= 50 \times 4.2124 + 1,000 \times 0.74726 = 210.62 + 747.26 = \mathbf{€957.88}$$

Step 2: Apply Regula Falsi.

$$YTM \approx r_1 + (r_2 - r_1) \cdot \frac{P(r_1) - P_{\text{target}}}{P(r_1) - P(r_2)}$$

$$YTM \approx 5\% + 1\% \cdot \frac{1,000 - 980}{1,000 - 957.88} = 5\% + 1\% \cdot \frac{20}{42.12}$$

$$\boxed{YTM \approx 5\% + 0.475\% = \mathbf{5.475\%}}$$

Accept answers in the range **5.4% – 5.5%** depending on rounding of P(6%).

> **Common error:** Students sometimes set up the interpolation fraction upside-down. Teach the mnemonic: numerator = how far the *price* has moved from the lower-rate price to the target price; denominator = the total price distance between the two trial prices. The YTM moves proportionally.

Time: ~20 min

4.4.3 c) Stock Valuation — Gordon Growth Model

$$D_1 = D_0 \times (1 + g) = 2.50 \times 1.03 = \mathbf{€2.575}$$

$$P_0 = \frac{D_1}{r - g} = \frac{2.575}{0.08 - 0.03} = \frac{2.575}{0.05} = \boxed{\textbf{€51.50 per share}}$$

Current market price: **€55.00** → The stock is **overvalued** by **€3.50 per share** (6.8% above intrinsic value).

For the full holding: 400 shares × €3.50 = **€1,400 overvaluation**.

> **Common error (D_0 trap):** Students frequently use D_0 = €2.50 in the numerator instead of D_1 = €2.575. Emphasise: the model prices the stock at $t = 0$ as the present value of *future* dividends — the first future dividend is D_1, which is one period of growth beyond the *last paid* dividend. Using D_0 undervalues the stock by the factor (1+g).

> **Discussion prompt:** "Would you sell the shares?" There is no one right answer — the model assumes constant growth forever, which is a simplification. A student who argues market expectations have revised g upward deserves credit. The point is to interrogate the model's assumptions, not just calculate.

Time: ~20 min

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5 Discussion Plan (Time Budget: 180 Minutes)

Phase	Content	Duration
Opening	Distribute case; students read independently. Brief framing: "Four envelopes, four tools."	10 min
Aufgabe 1	FV calculation → Fisher equation (both forms) → algebraic rate isolation	35 min
Aufgabe 2	PVIFA derivation + calculation → investment decision → CRF → perpetuity	40 min
Aufgabe 3	Three amortization tables → comparison table → recommendation with liquidity argument	50 min
Aufgabe 4	Bond price (discount/premium) → YTM interpolation → Gordon Growth Model	35 min
Synthesis	Instructor closes: all four tools are the same engine (PV/FV), applied to different cash flow shapes	10 min
Total		**180 min**

Buffer note: Aufgabe 3 is the longest and most mechanical. If time is short, assign Option C (Bullet) as read-only rather than a computed table — students state the cash flows verbally. This saves ~5 minutes.

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6 Board Plan

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LEFT PANEL — Formulas                    RIGHT PANEL — Results (built up live)
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Ch.1:  FV = PV × (1+i)^n                A1a) FV = €608,326
       r = i_nom − π (approx)             A1b) r_real ≈ −1.5% / exact −1.42%
       r = (1+i)/(1+π) − 1 (exact)      A1c) i_req = 5.77%

Ch.2:  PV = C × PVIFA(i,n)              A2a) PV = €932,376
       PVIFA = [1−(1+i)^−n] / i         A2b) Worth buying ✓ (€32k surplus)
       C = PV × CRF                       A2c) Mortgage = €82,370/yr
       PV_∞ = C / i                       A2d) Perpetuity = €1,600,000

Ch.3:  Scissors Effect diagram            A3 Comparison table (fill in live)

Ch.4:  P = C×PVIFA + F×(1+r)^−n        A4a) Bond = €901.65 (discount)
       YTM via interpolation              A4b) YTM ≈ 5.475%
       P₀ = D₁ / (r − g)               A4c) Fair = €51.50 → overvalued €3.50
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7 Instructor Alerts

1. **The Fisher sign trap (A1b):** Students almost universally get the sign direction wrong on first attempt. Before they calculate, ask: "Given 10% inflation and 4% nominal — are you getting richer or poorer?" The intuition must precede the formula.
2. **The D_0/D_1 distinction (A4c):** This is the single most common DDM error. Make it visceral: "Georg *already received* €2.50. That's gone. What will he receive *next year*? That's what you're buying." The stock price is the PV of what comes *after* today.
3. **Annuity table rounding errors (A3):** When students compound rounding across 4 rows, the closing balance in Year 4 rarely hits exactly zero — it may be ±€1. Accept ±€2 as correct. More importantly, use this moment to demonstrate why professional models carry more decimal places.

4. **Bullet loan liquidity appeal vs. total cost (A3d):** Students from finance backgrounds will often correctly identify Option C as the right choice for Year 1 liquidity. Students from accounting backgrounds may fixate on the €48k total interest as “expensive” and recommend Option B. Both positions are defensible if argued correctly — the goal is to see the trade-off, not to converge on one answer.